

ATLAS PIHAT

Pages 1 through 13 go over how audio is handled by the PiHat via the WM8960 Module, the SA818, and the ADS7830. Please verify these things are reflected exactly by the existing PCB. If they are not, recommend corrections.

Pages 14 through 26 go over how power is handled by the PiHat via TPS22918DBVR, FS8205A, DW01A, IP5306, LM66200DRLR, and the ADS7830. The peak output allowable is currently 2.4a peak. That is not sufficient power for the Pi5 and the PiHat accessories.

1. What the Audio System Must Do

The HAT must support the following operations using **only** software (I²C register writes to the WM8960 codec, plus the PTT GPIO for keying the SA818). No physical audio switches, no GPIO-controlled analog switch ICs.

1. **Radio receive** — incoming FM signal from the SA818 is demodulated to audio. The audio plays through the speaker so the operator can hear it, and is simultaneously captured by the Pi via I²S for any digital processing.
2. **Radio transmit, Pi-generated audio** — the Pi generates audio digitally, the codec converts it to analog, and the SA818 modulates it onto FM. PTT is asserted by software, no operator action needed. Used for any application where the Pi is the audio source.
3. **Radio transmit, microphone** — operator speaks into the mic and presses the PTT button; the codec switches to the mic input via I²C, and audio passes through to the SA818 for FM modulation.
4. **Local playback** — the Pi plays back audio (UI sounds, alerts, recordings, any other content) through the speaker without keying the radio.
5. **External audio input (aux in)** — operator plugs an external source (recorder, phone, another radio) into the aux input. The SA818 receive audio is automatically disconnected. External audio is captured by the Pi and also plays through the speaker.
6. **External audio output (aux out)** — whatever the codec is producing is always available on the aux output jack for headphones, external recording, or feeding another device. No switching, always live.

The speaker handles **all** audible output. Speaker volume is set by an external panel-mount potentiometer wired to the PCB via a JST connector; the pot's wiper is read through the ADS7830 and applied by software to the codec's speaker volume register — operator-

friendly tactile control without putting a pot in the audio signal path. The aux output is independent and always live. The mic is hardwired and selected by the codec via I²C when needed.

2. Architecture Summary

The WM8960 audio codec is the central routing engine. Three audio sources connect to three separate WM8960 left-channel inputs (LINPUT1, LINPUT2, LINPUT3). The codec's internal input mux (controlled by I²C) selects which source feeds the ADC and the output mixers. No external analog switches are required.

A switched 3.5mm TRS jack on the aux input handles SA818 / external-source selection **mechanically** — when nothing is plugged in, a normally-closed contact routes SA818 audio to LINPUT1; when a cable is inserted, the contact breaks and external audio routes to LINPUT1 instead. This single mechanical switch is the only physical routing element in the audio system.

Speaker output uses the WM8960's integrated differential Class-D speaker amplifier (SPK_LP / SPK_LN). No external amplifier IC.

The aux output uses the WM8960 headphone driver (HP_L, HP_R). The same HP_L signal also drives the SA818 transmit audio input through a fixed resistor divider — the SA818 receives whatever audio is currently being output, and only transmits when PTT is asserted.

All audio jacks are **off-board** panel-mount components wired to the PCB via JST connectors. **No 3.5mm jacks shall be PCB-mounted.**

3. WM8960 Pin Assignments — Authoritative

Pin	Name	Function	Connection
1	MICBIAS	Microphone bias output (~2.5V)	Through 2.2kΩ resistor to MIC_BIASED net (to mic JST)
2	LINPUT3 / JD2	Unused	Tie to AGND through 10kΩ to prevent floating
3	LINPUT2	Microphone signal input	AC-coupled (1μF) from MIC_BIASED net

Pin	Name	Function	Connection
4	LINPUT1	SA818 RX / aux input (mux via switched jack)	AC-coupled (1 μ F) from AUX_COM net
5	RINPUT1	Unused	Tie to AGND through 10k Ω
6	RINPUT2	Unused	Tie to AGND through 10k Ω
7	RINPUT3 / JD3	Unused	Tie to AGND through 10k Ω
8	DCVDD	Digital core supply	3.3V, 100nF decoupling
9	DGND	Digital ground	Ground plane
10	DBVDD	Digital buffer supply	3.3V, 100nF decoupling
11	MCLK	Master clock input	12.288 MHz from OSC1
12	BCLK	I ² S bit clock	GPIO 18 (Pi)
13	DACLRC	I ² S DAC frame clock	GPIO 19 (Pi)
14	DACDAT	I ² S DAC data	GPIO 21 (Pi)
15	ADCLRC	I ² S ADC frame clock	GPIO 19 (Pi) — shared with DACLRC
16	ADC DAT	I ² S ADC data	GPIO 20 (Pi)
17	SCLK	I ² C clock	GPIO 3 (Pi), 4.7k Ω pull-up
18	SDIN	I ² C data	GPIO 2 (Pi), 4.7k Ω pull-up
19	SPK_RN	Right speaker negative	Unused, leave open
20	SPKGND2	Right speaker ground	Ground plane
21	SPKVDD2	Right speaker supply	5V, 10 μ F + 100nF decoupling
22	SPK_RP	Right speaker positive	Unused, leave open
23	SPK_LN	Left speaker negative	SPEAKER JST pin 2
24	SPKGND1	Left speaker ground	Ground plane

Pin	Name	Function	Connection
25	SPK_LP	Left speaker positive	SPEAKER JST pin 1
26	SPKVDD1	Left speaker supply	5V, 10 μ F + 100nF decoupling
27	VMID	Mid-rail reference	1 μ F + 100nF to AGND, no other connection
28	AGND	Analog ground	Ground plane (star-grounded to DGND at single point)
29	HP_R	Headphone right output	AUX_OUT JST pin 2 (capless mode — no DC block cap)
30	OUT3	Auxiliary output / capless HP reference	AUX_OUT JST pin 3 (ground reference for capless HP)
31	HP_L	Headphone left output	AUX_OUT JST pin 1 AND TX audio path to SA818 (see §6)
32	AVDD	Analog supply	3.3V via ferrite bead, 10 μ F + 100nF decoupling
33	GND (pad)	Exposed pad	Ground plane

Capless headphone mode is required. Configure the WM8960 to use OUT3 as the headphone return reference (register R28 bit OUT3=1). This eliminates the need for large (220 μ F+) DC-blocking caps on HP_L and HP_R, saving board space and improving low-frequency response.

4. Input Signal Path — Mic (LINPUT2)

The microphone is **always wired** to LINPUT2. The codec selects whether to read it via I²C (mux register).

Bias network:

- WM8960 MICBIAS (pin 1) outputs a clean, regulated bias voltage (~2.5V with default register setting).
- A 2.2k Ω resistor from MICBIAS to the MIC_BIASED net supplies bias current to the electret capsule.

- A 100nF capacitor from MIC_BIASED to AGND decouples bias noise.
- The MIC_BIASED net is AC-coupled to LINPUT2 through a 1μF capacitor — this blocks the DC bias from reaching the codec input pin while passing audio.

Do not source mic bias from the battery (B+) or any other rail. Use the WM8960 MICBIAS pin exclusively. The current schematic's use of R27 (2.2kΩ from B+) is incorrect and must be changed.

ESD protection on the MIC_IN JST: TVS diode array between signal and GND, rated for ~5V working voltage.

5. Input Signal Path — SA818 RX / External Aux In (LINPUT1, switched)

LINPUT1 receives **either** SA818 receive audio (default) **or** external aux audio (when a cable is plugged into the off-board aux input jack). The selection is mechanical, automatic, and requires no GPIO or I²C action.

SA818 AF_OUT coupling:

- SA818 pin 3 (AF_OUT) is DC-coupled with internal bias.
- A 1μF capacitor (C19 in current schematic — retain) AC-couples AF_OUT to the SA818_AUDIO net.
- No attenuator is needed; AF_OUT delivers ~1Vpp at full software volume, which is within LINPUT1's input range with the PGA at unity.

External aux input:

- The off-board 3.5mm jack is a **switched TRS type**, with an NC (normally-closed) contact on the Tip.
- The Tip terminal of the jack wires to the AUX_COM net (the common signal that reaches LINPUT1).
- The NC contact of the jack wires to the SA818_AUDIO net (the AC-coupled SA818 output).
- When no plug is inserted: NC contact closes → SA818_AUDIO connects to AUX_COM → LINPUT1 receives SA818 audio.
- When a plug is inserted: NC contact opens → SA818 is disconnected → Tip carries external audio to AUX_COM → LINPUT1 receives external audio.

LINPUT1 AC coupling:

- The AUX_COM net is AC-coupled to LINPUT1 (pin 4) through a 1μF capacitor.

- This blocks any DC offset from external sources and the (already-coupled) SA818 path.

ESD protection on the AUX_IN JST: TVS diode array on the external Tip line.

6. Output Signal Paths

The WM8960 produces three independent audio outputs in this design:

6.1 Speaker — SPK_LP / SPK_LN (differential Class-D)

- WM8960 pins 25 and 23 drive an external speaker directly through the SPEAKER JST (2-pin).
- The WM8960 internal Class-D amplifier produces ~1W into 8Ω or ~1.5W into 4Ω.
- **Speaker impedance: 8Ω** (specified). 4Ω may be substituted, but the output filter design must change accordingly.
- For 8Ω speakers in a sealed plastic enclosure, **filterless operation is acceptable** — no external LC filter is required. The differential output and twisted-pair speaker cable provide adequate EMI containment.
- If the enclosure or compliance testing later requires filtering, add 22μH + 1μF per leg (low-pass at ~33kHz) close to the WM8960.

Volume control. Speaker volume is set by an external panel-mount potentiometer wired to the PCB via a 3-pin JST (see §8.5). The pot is a simple voltage divider — 3.3V on one end, ground on the other, wiper to the ADS7830. Software polls the ADS7830 at 20–50 Hz and maps the wiper position to the WM8960 SPKLVOL / SPKRVOL registers (R26/R27, range 0–63 covering -73dB to +6dB in 1dB steps).

- **Recommended pot:** 10kΩ linear or audio-taper, single-turn panel mount.
- **Current draw:** 330μA worst case (10kΩ across 3.3V). Negligible.
- **No effect on audio path.** The pot carries microamps from the 3.3V rail; the speaker audio remains a clean differential Class-D drive directly from the WM8960 to the speaker JST. Pot failure (open or short) cannot damage the audio path or change the speaker drive impedance.
- **Software taper.** With an audio-taper pot, software can use a simple linear mapping ($\text{SPKLVOL} = \text{ADC_value} \gg 2$). With a linear pot, software should apply a logarithmic curve to match human hearing perception.
- **Hysteresis recommended** (~2 LSB) in software to prevent register jitter when the pot is at rest.

The rotary encoder is NOT used for volume in this design — it remains available for menu navigation, frequency tuning, and other operator actions.

6.2 Aux Output — HP_L / HP_R (capless)

- WM8960 pins 31 (HP_L) and 29 (HP_R) drive the off-board aux output jack through the AUX_OUT JST.
- **Capless mode is mandatory** — OUT3 (pin 30) serves as the headphone return reference, eliminating large output capacitors.
- The aux output is **always live**, independent of which operating mode is active. Whatever the codec is producing goes out the aux output.
- A 47Ω series resistor on HP_L (and on HP_R if stereo aux out is wired) provides short-circuit protection. Already present as R20 — retain for HP_L, mirror for HP_R.
- ESD protection on the AUX_OUT JST.

HP volume is controlled via I²C (registers R2/R3: LOUT1VOL / ROUT1VOL).

6.3 SA818 Transmit Audio — HP_L through resistor divider

The SA818 MIC_IN pin (pin 18 of SA818) requires $\sim 100\text{mV}_{\text{pp}}$ for full FM deviation ($\sim \pm 5\text{kHz}$). The WM8960 HP_L outputs up to $\sim 1\text{V}_{\text{pp}}$. A fixed resistor divider attenuates $\sim 10:1$.

Divider network (codec side → SA818 side):

- Series resistor R_top: $9.1\text{k}\Omega$ (1% tolerance)
- Shunt resistor R_bot: $1\text{k}\Omega$ (1% tolerance)
- Series DC-blocking capacitor: $1\mu\text{F}$ (between divider output and SA818 MIC_IN)
- Optional: $10\text{k}\Omega$ trimmer in parallel with R_bot for field calibration (footprint only, populated if needed)

This divider taps off HP_L — it does **not** require a separate codec output. The same HP_L signal goes to both the aux output jack (for monitoring) and the SA818 (for transmit). During transmit, the operator hears what's being transmitted on the aux output. During receive or playback, HP_L carries received audio or Pi-generated audio — the SA818 ignores it unless PTT is asserted.

The $9.1\text{k}\Omega$ resistor presents a high impedance to HP_L, not loading the headphone driver meaningfully. The divider operates correctly whether or not headphones are plugged into the aux output.

7. Operating Mode — Signal Flow Reference

For each mode, the table shows the signal path and the I²C / GPIO state required. **PTT** (GPIO 17 HIGH = TX keyed) is the only GPIO used for audio mode control.

Mode	Source → LINPUTx (mux)	Output mixer routing	PTT	Notes
Radio RX	SA818 → LINPUT1 (jack NC)	LMIXER → SPK_LP/LN (speaker) AND HP_L (aux out)	LOW	I ² S ADC always captures LINPUT1 — Pi can process or record the audio simultaneously
TX (Pi audio)	(none — Pi generates)	DAC → HP_L → R-divider → SA818; also HP_L → aux out monitor; mute SPK_LP/LN to avoid acoustic coupling	HIGH	Pi streams audio to I ² S DAC
TX (mic)	Mic → LINPUT2 (codec switches mux on PTT)	LMIXER → HP_L → R- divider → SA818; mute SPK_LP/LN	HIGH	Low-latency analog path through codec mixer, no Pi involvement required
Playback	(none — Pi generates)	DAC → SPK_LP/LN (speaker); DAC → HP_L (aux out)	LOW	SA818 receives audio but is not keyed, so no transmission
Aux RX	External → LINPUT1 (jack TIP)	LMIXER → HP_L AND SPK_LP/LN	LOW	NC contact opens automatically when cable inserted

Software state machine summary:

- On `(PTT_PRESSED)` event: (a) write WM8960 mux register to route LINPUT2 (mic) to LADC instead of LINPUT1; (b) configure mixer for analog passthrough mic → HP_L; (c) mute SPK_LP/LN; (d) drive GPIO 17 HIGH to key SA818.
- On `(PTT_RELEASED)` event: reverse the above — drop GPIO 17 LOW, restore SPK_LP/LN, switch mux back to LINPUT1.
- On `(PI_TX_REQUEST)` event (software-initiated transmit, e.g. for Pi-generated audio): drive GPIO 17 HIGH, stream audio via I²S DAC, drop GPIO 17 LOW on completion.

8. Off-Board Jack Wiring — JST Pinouts

All audio connectors on the PCB are JST headers. The 3.5mm panel-mount jacks are off-board, wired via short flying leads (≤ 15 cm). Use shielded cable for runs longer than 10 cm or in noisy environments. **No 3.5mm jacks on the PCB.**

8.1 AUX_IN JST — 4-pin (XH or PH series, vertical or right-angle)

Pin	Net	Description	Wire to off-board jack
1	AUX_COM	Common signal to LINPUT1	Jack Tip terminal
2	SA818_AUDIO	AC-coupled SA818 AF_OUT	Jack Tip-NC (normally-closed) terminal
3	GND	Audio ground	Jack Sleeve terminal
4	NC (no connect)	Reserved	Leave unconnected

Jack type: Switched 3.5mm TRS mono with NC contact on Tip. Recommended: CUI SJ-3523N or equivalent (3 contacts: Tip, Tip-NC-spring, Sleeve).

8.2 AUX_OUT JST — 3-pin

Pin	Net	Description	Wire to off-board jack
1	HP_L	Left output (via 47Ω)	Jack Tip terminal
2	HP_R	Right output (via 47Ω)	Jack Ring terminal
3	GND	Audio ground (OUT3 return for capless)	Jack Sleeve terminal

Jack type: Standard 3.5mm TRS stereo (non-switched). The aux out is always live; no insertion-detect is needed.

8.3 MIC_IN JST — 2-pin

Pin	Net	Description	Wire to off-board jack
1	MIC_BIASED	Mic signal + bias (single wire)	Jack Tip terminal
2	GND	Mic ground	Jack Sleeve terminal

Jack type: Standard 3.5mm TS (mono) or TRS with Ring tied to Sleeve. Electret capsule wires Tip-to-Sleeve at the mic end.

8.4 SPEAKER JST — 2-pin

Pin	Net	Description
1	SPK_LP	WM8960 SPK_LP (positive)
2	SPK_LN	WM8960 SPK_LN (negative)

Connector type: JST PH 2.0mm or XH 2.5mm, 2-pin, polarized. Speaker: 8 Ω , 1–3W rating, twisted-pair leads recommended.

8.5 VOLUME_POT JST — 3-pin

For the off-board speaker volume potentiometer. The pot is mounted on the enclosure panel; this JST connects it to the PCB.

Pin	Net	Description	Wire to off-board pot
1	3V3	3.3V supply (one end of pot resistive track)	Pot CCW terminal
2	VOL_WIPER	Wiper signal $\rightarrow$ ADS7830 CH1 (through R9, 1k Ω series)	Pot wiper (centre terminal)
3	GND	Ground (other end of pot resistive track)	Pot CW terminal

Pot specification: 10k Ω $\pm$ 20%, single-turn, panel-mount, 6mm shaft. Linear or audio-taper acceptable (software handles the curve). Recommended part: Bourns PDB181-K420K-103B (10k Ω linear) or similar.

Connector type: JST PH 2.0mm or XH 2.5mm, 3-pin, polarized.

Silkscreen label: VOL POT and pin numbers, with 3V3 / WIPER / GND notation.

8.6 ADC_BREAKOUT JST — 8-pin

The remaining 6 ADS7830 channels are broken out to a single 8-pin JST connector for external sensors, resistor trees, switches, or other analog inputs the operator wires up later.

Pin	Net	Description
1	3V3	3.3V supply for external resistor trees, sensor reference, etc. ($\leq 100\text{mA}$ from Pi 3V3 rail)
2	ADC2	ADS7830 CH2 input, with R24 (1k Ω) series protection
3	ADC3	ADS7830 CH3 input, with R25 (1k Ω) series protection
4	ADC4	ADS7830 CH4 input, with R26 (1k Ω) series protection
5	ADC5	ADS7830 CH5 input, with R28 (1k Ω) series protection
6	ADC6	ADS7830 CH6 input, with R29 (1k Ω) series protection
7	ADC7	ADS7830 CH7 input, with R30 (1k Ω) series protection
8	GND	Ground reference for external circuits

ADS7830 channel allocation summary:

- CH0 — Battery voltage divider (committed, on-board)
- CH1 — Speaker volume pot (committed, via VOLUME_POT JST §8.5)
- CH2-CH7 — General-purpose analog inputs (broken out via this JST)

Connector type: JST PH 2.0mm or XH 2.5mm, 8-pin, polarized. PH is preferred for the smaller footprint given the board's space constraints.

Silkscreen label: ADC with pin numbers and one of 3V3 / 2 / 3 / 4 / 5 / 6 / 7 / GND per pin. Each pin's channel number (2–7) must be clearly readable so the operator can identify which channel they're using.

Input range: Each ADC channel accepts 0V to 2.5V (the ADS7830's internal reference). Inputs above 2.5V will be clipped at full-scale. Inputs above 3.6V risk damaging the ADS7830 input; the 1k Ω series resistor provides limited protection — for safety-critical external circuits, add a clamp diode at the sensor end.

9. PCB Layout Requirements (Audio)

1. **Star ground topology.** AGND and DGND of the WM8960 join at a single point near the codec's exposed pad. The AGND zone covers the area under the codec, the input

AC-coupling caps, and the SA818 audio coupling cap. DGND covers the area under I²S and I²C traces.

2. **Ferrite bead** between the 3.3V digital supply and the WM8960 AVDD pin (pin 32). Required for analog supply cleanliness.
 3. **Coupling capacitor placement.** All audio AC-coupling caps (1μF film or C0G ceramic; avoid X7R for signal coupling due to DC bias distortion) within 5mm of their associated WM8960 pin.
 4. **Audio trace routing.** Audio signal traces (SA818_AUDIO, AUX_COM, MIC_BIASED, HP_L, HP_R, SPK_LP, SPK_LN) shall not run parallel to or under: I²S clocks (BCLK, LRCLK, DACDAT, ADCDAT), UART traces, the boost converter switching node, or the SA818 RF section. Cross at 90° when crossings are unavoidable.
 5. **R-divider placement** (for SA818 TX audio) close to the SA818 MIC_IN pin (pin 18), not close to HP_L. This minimizes the trace length carrying the attenuated low-level signal, which is the most noise-sensitive segment.
 6. **MICBIAS routing.** The MICBIAS net (codec pin 1 to the 2.2kΩ bias resistor) runs only on the analog ground side. Decoupling cap (100nF) within 3mm of the bias output.
 7. **SA818 RF isolation.** No audio traces within 3mm of the SA818 antenna trace or the SMA connector. The SA818 ground pad is continuous with the AGND zone through the codec's analog ground area.
 8. **JST connector placement.** Audio JST connectors along one edge of the board, away from the SA818 RF section and the boost converter. Allows clean cable routing to the enclosure panel.
 9. **MCLK crystal/oscillator** (12.288 MHz) within 10mm of the WM8960 MCLK pin (pin 11). Treat the MCLK trace as a clock — guarded by ground on both sides if it runs longer than 20mm.
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10. I²C Register Configuration (Reference)

The following WM8960 registers govern the audio routing. The PCB designer does not need to implement this — it's listed for completeness and to confirm that the hardware architecture supports the required modes.

Register	Function	Notes for this design
R0 (LINVOL)	LINPUT1/2/3 PGA volume L	Software sets per mode
R15 (RESET)	Soft reset	Always first
R17 (CLOCKING1)	MCLK divider	12.288 MHz input
R19 (POWER MGMT 1)	ADC, MICB, VMID power	Enable VMID, MICBIAS, AINL
R20 (POWER MGMT 2)	DAC, output, LMIX power	Enable LOUT1 (HP_L), SPK_LP, LMIX, DACL
R22 (LEFT OUT MIX)	LMIX source	Select LI2LO (LINPUT bypass) for mic-to-radio analog path; DACL for Pi-generated audio TX or playback
R26 / R27 (SPKLVOL / SPKRVOL)	Speaker volume L / R	Software writes from pot wiper position (ADS7830 CH1 read)
R28 (ADDITIONAL CTL 4)	OUT3 enable	Bit OUT3=1 for capless mode
R32 (LEFT ADC INPUT SIG)	LADC input select	INPUT1 for RX modes; INPUT2 for mic TX
R34 (RIGHT ADC INPUT SIG)	RADC input select	Disabled (right channel unused)
R37 (LEFT ADC SIGNAL PATH)	LMIC boost	Configure for line input vs mic gain
R47 (POWER MGMT 3)	LMIC, RMIC, LOMIX, ROMIX	Enable LMIC, LOMIX
R51 (CLASS D CONTROL 1)	SPKL/R enable	Bit SPK_OP_EN_L=1
R52 (CLASS D CONTROL 3)	Class D level	Configure for speaker volume range

ATLAS HAT — Power Architecture Specification

Project: ATLAS Transceiver — Custom Raspberry Pi HAT **Document type:** Description of the power system as currently implemented, plus open items requiring evaluation **Scope:** Power generation, distribution, switching, and monitoring. Audio routing is specified in a separate document. **Target Pi:** Raspberry Pi 5 (note: earlier revisions targeted Pi 4B — power numbers have been updated)

This document describes how power is handled on the ATLAS HAT as currently designed in [Schematic_Jason_atlas_pi_hat_2026-05-17.pdf](#). Several items at the end require the reviewer's evaluation and recommendation, most notably battery selection and whether the current boost converter has adequate headroom for the Pi 5.

1. What the Power System Must Do

1. **Power the Pi 5 and all HAT peripherals** from a single internal lithium battery, with USB-C charging.
 2. **Charge the battery and run the device simultaneously**, with no interruption when the charger is plugged in or unplugged.
 3. **Survive a charger disconnect mid-operation** with zero glitch on the 5V rail. The Pi 5 must not see any dip that would cause a reboot or filesystem corruption.
 4. **Allow the operator to power the main device off** via a panel switch, while leaving the LoRa subsystem powered and operational.
 5. **Allow the operator to override the LoRa power** independently — turn LoRa off via a dedicated panel switch when needed.
 6. **Allow software to power down the SA818 radio** to save battery when only LoRa or GPS is needed.
 7. **Report battery state of charge** to the Pi so it can display a battery indicator and warn before shutdown.
 8. **Provide 5V power to the off-board 3-inch LCD screen** via a dedicated JST connector. The LCD is USB-powered (5V + GND only) and uses SPI from the Pi directly for data.
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2. Architecture Summary

The power chain is built around four ICs:

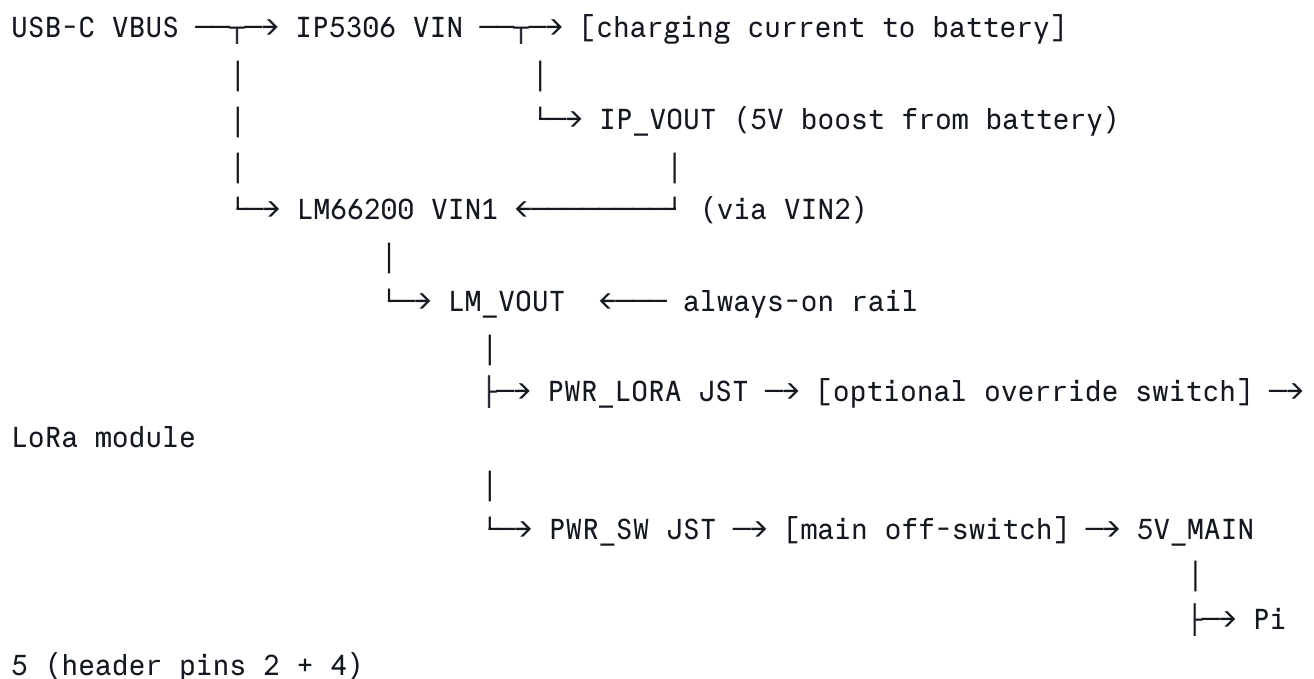
- **DW01A + FS8205A** — single-cell lithium battery protection (overcharge / overdischarge / overcurrent).
- **IP5306** — combined charger and 5V boost converter. Charges the battery from USB-C VBUS, boosts battery voltage to a regulated 5V output rail.
- **LM66200** — dual-input ideal-diode power MUX. Selects automatically between USB-C VBUS and the IP5306 boost output, providing glitch-free switching when the charger is plugged or unplugged. This is what gives the system "no interruption on charge state change."
- **TPS22918** — load switch with GPIO enable, used to power the SA818 radio on demand under software control.

Battery voltage is monitored via a resistor divider into ADS7830 channel 0 (the same ADC used for the audio volume pot and the breakout). Software calculates state of charge from the voltage reading.

Two external panel-mount switches are wired to the HAT via JST connectors:

- **PWR_SW** — main device on/off. Cuts power to the Pi 5, the SA818 rail, the LCD, and the WM8960 codec. Does NOT cut power to LoRa.
- **PWR_LORA** — LoRa override. Cuts power specifically to the LoRa module. Independent of PWR_SW.

3. Power Tree



TPS22918 → SA_5V → SA818



LCD_PWR JST → 3" LCD



WM8960 SPKVDD1/2 (5V analog)



Battery → DW01A + FS8205A protection → B+ → IP5306 BAT pin

B+ → R34/R35 divider + C1 filter → ADS7830 CH0 (fuel gauge)

4. Components and Their Roles

4.1 DW01A + FS8205A — Battery Protection (U4, Q2)

Standard single-cell lithium protection circuit. Connects directly to the battery JST and to the IP5306 BAT pin.

- **Overcharge protection:** Disconnects battery at ~4.25V.
- **Overdischarge protection:** Disconnects battery at ~2.4V. (Note: this is below the safe operating range for the Pi 5 — software should initiate clean shutdown well before this point. See §8.)
- **Overcurrent protection:** Trips at ~3A discharge (set by FS8205A Rds(on)).
- **Short circuit protection:** Fast trip on dead-short conditions.

4.2 IP5306 — Charger + 5V Boost (U3)

Combined battery management and boost converter IC. Pulls input from USB-C VBUS, charges the battery, and simultaneously boosts the battery voltage to a regulated 5V output (IP_VOUT) for system use.

- **Input voltage:** 5V from USB-C VBUS.
- **Charge current:** Up to 2A (typical) into the battery.
- **Boost output:** 5V ±5%, rated **2.4A peak** on the standard IP5306 part.
- **Variant in use:** Standard IP5306 (no I²C interface). Confirmed not the IP5306-I2C variant.
- **Indicator LEDs:** LED1, LED2, LED3 outputs drive three on-board status LEDs (via R31, R32, R33 1kΩ current-limit resistors) — display charge state and battery level.
- **Inductor:** L1, 1μH, on the switching node.
- **Bulk capacitance:** C21 (1000μF) on output, C14 (10μF) and C22 (10μF) on input.

4.3 LM66200 — Power Path Selector (U1)

Dual-input ideal-diode power MUX. The component that makes "charge while running" work.

- **VIN1:** USB-C VBUS (direct).
- **VIN2:** IP5306 boost output (IP_VOUT).
- **VOUT:** LM_VOUT (always-on rail feeding everything downstream).
- **Selection logic:** Automatically passes the higher-voltage input to the output. When USB-C is plugged in (5V VBUS), it dominates. When unplugged, the boost output (~5V from battery) takes over.
- **Glitch-free switching:** This is the key feature. The Pi 5 sees a continuous 5V supply regardless of charger state.

4.4 TPS22918 — SA818 Load Switch (U5)

Load switch with logic-level enable for software-controlled SA818 radio power.

- **VIN:** 5V_MAIN.
- **VOUT:** SA_5V (feeds SA818 VBAT pin).
- **Enable (ON pin):** Driven from Pi GPIO 22 via R1 (1kΩ). HIGH = SA818 powered, LOW = SA818 unpowered.
- **Current rating:** 2A continuous. SA818 peak draw is ~500mA, well within margin.
- **Built-in slew-rate control** prevents inrush spikes on the 5V_MAIN rail when SA818 turns on.
- **Flyback protection:** 1N5819 Schottky diode in the SA_5V path to handle inductive load behavior.

5. Rails and What They Feed

Rail	Voltage	Source	Always-on?	Feeds
B+	3.0–4.2V (battery range)	Battery via DW01A/FS8205A protection	Yes	IP5306 BAT pin; ADS7830 CH0 divider
VIN	5V	USB-C VBUS	When charger plugged in	IP5306 VIN; LM66200 VIN1

Rail	Voltage	Source	Always-on?	Feeds
IP_VOUT	5V (boosted)	IP5306 boost	Whenever battery is present	LM66200 VIN2
LM_VOUT	5V	LM66200 (auto-selects higher of VIN, IP_VOUT)	Yes — whenever battery OR USB-C is present	PWR_SW JST; PWR_LORA JST
5V_MAIN	5V	LM_VOUT through PWR_SW external switch	Only when main switch is ON	Pi 5; TPS22918 input; LCD; WM8960 5V supplies
SA_5V	5V	TPS22918 output (gated by GPIO 22)	Only when main switch is ON AND GPIO 22 HIGH	SA818 VBAT
3V3	3.3V	Generated by the Pi 5's onboard regulator	Whenever Pi 5 is powered	All HAT 3.3V logic: WM8960 DCVDD/DBVDD/AVDD, ADS7830, pull-ups, GPS, LoRa, etc.

The 3.3V rail is **not generated on the HAT** — it comes from the Pi 5 via the 40-pin header (pins 1 and 17). The Pi 5's 3.3V rail can supply approximately 500mA total; HAT consumers (codec digital + ADS7830 + pull-ups + GPS + LoRa) must fit within this budget.

6. JST Connectors — Power

6.1 Battery JST — 2-pin (JST PH 2.0mm)

- Pin 1: BAT+ (red wire convention)
- Pin 2: BAT− (black wire convention)
- Polarized — keying prevents reverse insertion, but silkscreen polarity marking required as backup.

6.2 USB-C — Charging Input

- USB-C receptacle, surface-mount, board-edge placement.
- CC1, CC2 each pulled to GND via 5.1k Ω (R12, R37) — declares the HAT as a sink-only USB-C device.
- Accepts standard 5V from any USB-C charger or power bank. Maximum input current is determined by the source (typically 1.5A from a basic charger, up to 3A from a compliant 15W source).
- **No USB-PD negotiation** — does not request higher voltages.
- **No data lines connected** — power-only role.

6.3 PWR_SW JST — 2-pin (main off-switch)

External panel-mount switch wired here.

Pin	Net	Description
1	LM_VOUT	Upstream side (always-on rail)
2	5V_MAIN	Downstream side (Pi 5, LCD, SA818 path, codec)

Switch closed = device ON. Switch open = device OFF (LoRa still runs if PWR_LORA is closed).

Switch specification: Panel-mount SPST toggle or slide switch, rated 5V / 3A minimum. Latching action (not momentary).

6.4 PWR_LORA JST — 2-pin (LoRa override switch)

External panel-mount switch wired here. Independent of the main PWR_SW.

Pin	Net	Description
1	LM_VOUT	Upstream (always-on rail)
2	LORA_VCC	Downstream (feeds the VCC pin of the LoRa 4-pin JST)

Switch closed = LoRa powered. Switch open = LoRa unpowered. Independent of main device state — LoRa can be on while the rest of the device is off, or off while the rest is on.

Switch specification: Same as PWR_SW. SPST toggle, panel-mount, latching.

Note: As currently drawn in the schematic, the LoRa module's VCC pin connects directly to LM_VOUT via the PWR_LORA pads — there is no switch in line by default. The schematic provides the tap point for an external switch; the operator wires it via the JST. **Confirm with the project owner whether they want this kept as-is (off-board switch required) or whether a logic-level enable circuit should be added on-board.**

6.5 LCD_PWR JST — 2-pin (new — currently absent from schematic)

For powering the off-board 3-inch round LCD screen.

Pin	Net	Description
1	5V_MAIN	5V supply to LCD
2	GND	Return

Connector type: JST PH 2.0mm, 2-pin, polarized.

Silkscreen: LCD PWR 5V / GND

Current budget: Assume up to 500mA for the LCD. Most 3-inch round IPS LCDs draw 200–400mA at 5V depending on backlight brightness. The LCD's data interface (SPI) connects directly to the Pi 5's SPI pins on the GPIO header — not to the HAT.

This connector is not currently on the schematic and must be added. See §10.

7. Software-Controlled Power Switching

7.1 SA818 Power Control (GPIO 22 → TPS22918)

The Pi can power the SA818 on or off in software by driving GPIO 22:

- GPIO 22 HIGH → TPS22918 ON → SA818 receives 5V → radio operational.
- GPIO 22 LOW → TPS22918 OFF → SA818 unpowered → ~500mA saved during battery-only operation.

The SA818 is also independently sleep-controllable via its PD pin (GPIO 27), but full power-down through the load switch eliminates the standby current entirely.

7.2 LoRa Power Control (none — manual switch only)

The LoRa module is not currently software-controllable. It is powered manually via the PWR_LORA external switch. This is by design — LoRa should remain operational even

when the Pi is off or has crashed.

If software control of LoRa power is later required (e.g., to save battery during deep sleep), a second load switch driven by a GPIO could be added between LM_VOUT and the LoRa JST. This is not currently in the schematic.

8. Battery Monitoring — ADS7830 Channel 0 (Fuel Gauge)

Battery voltage is measured by the on-board ADS7830 ADC for software fuel-gauge calculation.

- **Divider:** R34 (100k Ω) and R35 (100k Ω) form a 2:1 divider from B+ to ADS7830 channel 0.
 - **Filter:** C1 (100nF) shunt cap to GND provides RC low-pass filtering (~5ms time constant) to suppress noise.
 - **Scaling:** Battery range 3.0–4.2V $\rightarrow$ ADC sees 1.5–2.1V (well within the ADS7830's 0–2.5V input range).
 - **Quiescent draw from battery:** 200k Ω total divider impedance $\rightarrow$ 21 μ A at 4.2V $\rightarrow$ 0.5mAh per 24 hours. Negligible.
 - **Software fuel-gauge model:** Pi reads ADC value, applies a lookup table or polynomial to convert voltage to state-of-charge percentage. Should also include load-compensation if accurate SoC under varying load is required (Li-ion voltage sags significantly under heavy current draw — a reading taken during SA818 TX is not the same as a reading taken at idle).
 - **Recommended software shutdown threshold:** 3.2V battery voltage. Below this, the Pi 5 may brown out or the boost converter may struggle to maintain regulation. Hardware protection (DW01A) only kicks in at 2.4V, which is far too low for clean operation.
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9. Open Items for Reviewer Evaluation

9.1 IP5306 Boost Output — 2.4A peak may be insufficient

This is the most important item in this document. **The 2.4A peak boost output of the IP5306 must be evaluated against worst-case load.**

Worst-case load estimate, Pi 5 system:

Load	Current at 5V
Pi 5 under load (4 cores active, USB activity, etc.)	1.5-2.5A
Pi 5 burst peaks (very brief, e.g. SD card writes)	up to 5A
SA818 transmit	0.5A
3" LCD with backlight	0.3-0.5A
LoRa module TX burst	0.3A
WM8960 speaker at moderate volume	0.2A
Other (GPS, ADC, pull-ups, etc.)	0.1A
Sustained worst-case total	~3.0-3.5A
Burst worst-case total	>5A

The standard IP5306 is rated for 2.4A peak boost output. **Sustained operation under heavy Pi 5 load with SA818 transmitting and LCD on will exceed this rating.**

The Pi 4B was the original target (peak ~1.2A), and the IP5306 was sized for that platform. With Pi 5 the boost converter is the bottleneck.

Options for the reviewer to evaluate:

1. **Replace IP5306 with a higher-current boost solution.** Candidates: IP5389 (5A capable), TPS61288 + dedicated charger IC, or a Pi-Power-HAT-class solution. Higher cost, larger area, more components.
2. **Throttle the Pi 5.** Limit CPU clocks, undervolt, disable unused cores. Software solution, no hardware change. Reduces peak draw to ~1.5A sustained.
3. **Enforce load-shedding by software.** Never run SA818 TX while LCD backlight is at full brightness, etc. Lower system flexibility.
4. **Accept the limitation.** IP5306 will current-limit or drop voltage when overloaded. Pi 5 may brown out under combined heavy load. Acceptable only if heavy load conditions are rare and brief.

Recommendation requested from reviewer: which option (or combination) is appropriate for this design?

9.2 USB-C Source Compatibility with Pi 5 PMIC

The Pi 5 has a USB-PD-aware power management IC. When powered from a non-PD source providing only 5V/3A or less, the Pi 5 firmware may log under-voltage warnings or limit performance unless configured otherwise. Confirm:

- Whether the IP5306 boost output is sufficient to satisfy Pi 5's startup current draw without triggering brownout.
- Whether `usb_max_current_enable=1` in `/boot/firmware/config.txt` is required to disable Pi 5 PMIC current-limiting.

9.3 Battery Selection — Reviewer to Recommend

Required: Single-cell 3.7V lithium-ion or LiPo battery. **Maximum one cell. Do not parallel multiple cells.** The DW01A protection IC is designed for single-cell operation only.

Target operating time: 8 hours of typical use.

Typical-use load estimate:

Load	Average current at 5V
Pi 5 idle / light use	0.6A
3" LCD, backlight medium	0.3A
SA818 receive	0.05A
LoRa idle (mostly listening)	0.05A
Other	0.05A
Total at 5V	~1.05A
Equivalent at 3.7V (assuming 85% boost efficiency)	~1.7A

For 8 hours: $8h \times 1.7A = \sim 13.6Ah$ required battery capacity.

Reviewer should recommend a specific battery part meeting:

- Single-cell 3.7V lithium chemistry (Li-ion or LiPo)
- 12,000–15,000 mAh capacity
- JST PH 2.0mm connector
- Continuous discharge rating $\geq 3A$ (some pouch cells limit at 1C, which may be insufficient)

- Physical dimensions compatible with the intended enclosure (project owner to specify enclosure constraints)
- Available from a reputable supplier with consistent quality

9.4 Charge Time

At the IP5306's typical 2A charge rate, a 12,000 mAh battery will charge from empty to full in approximately **6-7 hours**. A 15,000 mAh battery: **7-8 hours**. Confirm with project owner whether this is acceptable; if faster charging is needed, the charger IC must change.

9.5 Thermal — IP5306 Sustained High Current

At 2A sustained boost output, the IP5306 dissipates 0.5-1W as heat. With a small SOP-8 package and minimal copper pour, the IC can reach 80°C+. Confirm:

- Thermal copper pour adequate on the PCB layout.
- Optional: small heatsink or thermal via array under the IC.

If §9.1 results in a different boost IC, re-evaluate thermal accordingly.

9.6 USB-C Input Overvoltage

The IP5306 input is rated to ~6V maximum. A non-compliant USB-C charger that applies higher voltage (some cheap PD chargers misbehave) will destroy the IC. Consider whether to add a TVS clamp or a Zener+fuse OVP circuit on the USB-C VBUS line. Project owner to decide based on intended user environment.

9.7 Quiescent Draw with "Device Off, LoRa On"

With the main PWR_SW open and PWR_LORA closed, the system should consume only:

- LM66200 quiescent: ~1μA
- IP5306 standby: ~150μA
- DW01A quiescent: ~3μA
- Battery sense divider: 21μA
- LoRa module idle: ~30mA (Meshtastic listening)
- **Total: ~30mA**

At 30mA on a 12,000 mAh battery, LoRa-only operation lasts ~**400 hours (~17 days)** without main power.

Reviewer should confirm that the LM66200 and IP5306 paths do not leak excessive current when the main switch is off.

10. Required Schematic Changes

1. **Add LCD_PWR JST** (2-pin, JST PH 2.0mm) connecting 5V_MAIN and GND. Silkscreen `LCD PWR 5V / GND`. See §6.5.
 2. **Confirm or formalize the LoRa override switch** at PWR_LORA. Currently the schematic taps LM_VOUT directly to the LoRa VCC pin via the PWR_LORA pads — adequate as a switch tap point, but should be silkscreened clearly: `LORA SW to panel switch`.
 3. **Silkscreen the main off-switch JST** clearly as `MAIN SW to panel switch`. (Current refdes is PWR_SW.)
 4. **Battery polarity silkscreen** at the battery JST (+ and - markings) as a backup to the connector keying.
 5. **Add a TVS diode** on USB-C VBUS for ESD protection (if not already present).
 6. **Evaluate and address the 2.4A boost limit** per §9.1 — this may require a different charger IC, in which case the schematic changes are substantial.
 7. **Update charge LED silkscreen** (LED1/LED2/LED3) with what each indicates — typically: red = charging, yellow = >50%, green = full.
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11. What the Power System Does NOT Do

- **No multi-cell battery support.** Single 3.7V cell only.
- **No parallel battery cells.** Use one larger cell, not two smaller ones in parallel — the protection IC and charger cannot safely manage parallel cells.
- **No USB-PD negotiation.** USB-C input is treated as a fixed 5V source.
- **No charge current adjustment.** IP5306 charge rate is fixed at its internal default (~2A).
- **No I²C-based fuel gauge IC.** Battery voltage is sampled by the general-purpose ADS7830 and software calculates state-of-charge. This is sufficient for a battery indicator but not as accurate as a dedicated Coulomb counter.
- **No coulomb counting.** State of charge is voltage-based only, which is moderately accurate at rest and less accurate under varying load. If precise SoC reporting is required, add an INA226 or similar current monitor.
- **No automatic LoRa power-down.** LoRa is on whenever PWR_LORA is closed, regardless of Pi state. If software-controlled LoRa is needed, hardware change required.

- **No solar input.** Charging is USB-C only.
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12. Deliverables Requested from Reviewer

1. **Recommended battery part number** meeting §9.3, with supplier link and datasheet.
 2. **Recommendation on 2.4A boost limit** (§9.1) — accept, replace, or mitigate, with reasoning.
 3. **Pi 5 USB-C compatibility confirmation** (§9.2) — what config.txt entries or firmware settings are needed.
 4. **Updated schematic** incorporating the changes in §10, plus any changes resulting from §9.1.
 5. **Updated BOM** with JLCPCB-available substitutes for any out-of-stock parts, especially if §9.1 results in a different boost IC.
 6. **Charge-curve calibration data points** for the chosen battery, suitable for software fuel-gauge implementation (similar to the PiSugar custom battery curve format).
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End of power architecture specification.